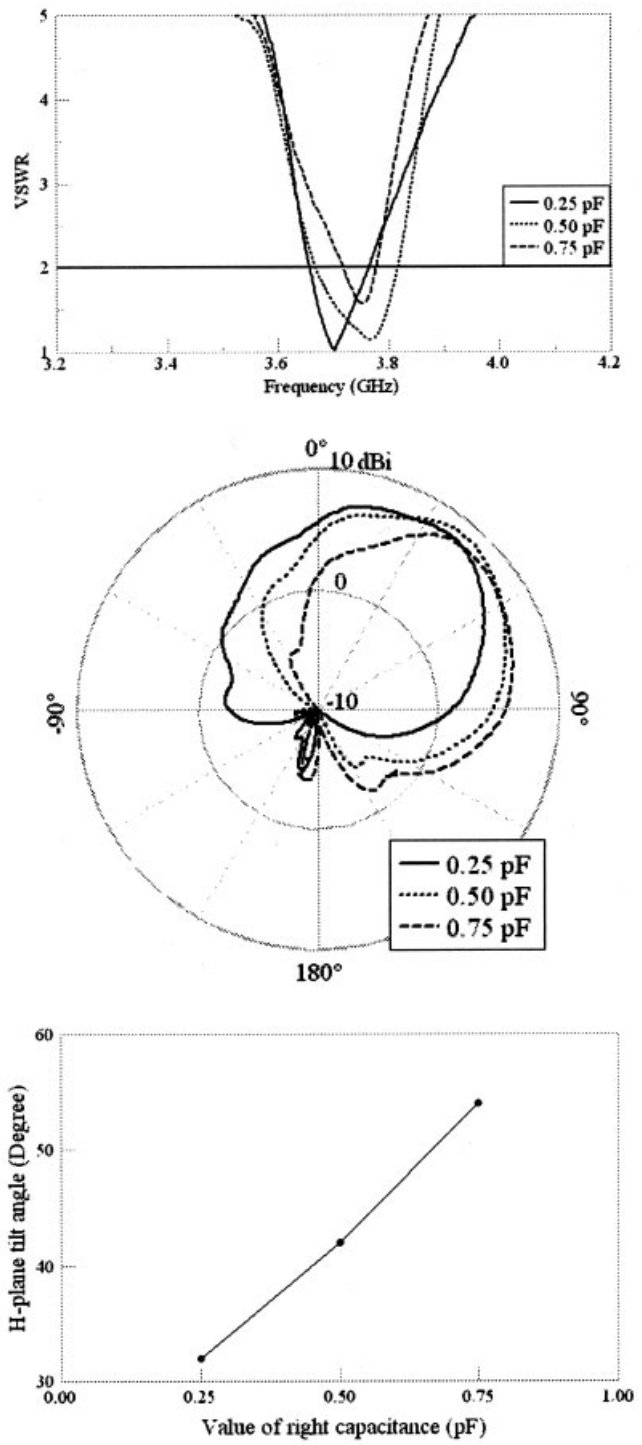


Figure 4 Measured shared VSWR bandwidth (top), resulting radiation patterns (middle), and tilt angle (bottom) for variation 3 of the original RMPA as the right capacitance is increased from 0.25 to 0.75 pF, the left capacitance set at 1.75 pF, and the inductances set at 1 nH

1 reconfigures its radiation direction in three switchable directions in the H-plane, that is, −25°, 0°, and +25°. In comparison with the original RMPA, it eliminates two switches and sacrifices only a slight decrease in the tilt angle of the pattern. After replacing the switches with varactors, variation 2 is able to continuously scan its radiation pattern from −27° and 27° by controlling the value of the varactors. Variation 3 expands the H-plane pattern scan range to −(32°, 54°) ∪ (32°, 54°) by using inductors to series-resonate the varactors in variation 2. Variations 1 and 2 have a combined impedance and pattern bandwidth of more than 4.5% with respect to the center frequency, while variation 3 has a bandwidth of only 1.5%. In all the experiments, copper strips were used as switches in an on/off state and differently valued chip capacitors were used to mimic the behavior of the varactors. Implementation of electronic control of the radiation behavior of these antennas using PIN diodes and varactors is underway.

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PRINTED CIRCULAR RING MONOPOLE ANTENNAS

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ABSTRACT: *This paper presents a novel design of printed circular-ring monopole antennas for ultra-wideband (UWB) applications. A circular-disc ring printed on a dielectric substrate and fed by a 50Ω microstrip line is demonstrated numerically and experimentally to achieve an ultra-wide 10-dB bandwidth with omnidirectional radiation patterns over the entire bandwidth. The design parameters and the performance of the proposed antenna are analyzed. Good agreement has been obtained between the simulation and the measurement.* © 2005 Wiley Periodicals, Inc. Microwave Opt Technol Lett 45: 372–375, 2005;
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Key words: *circular-ring monopole; microstripline-fed; ultra-wideband (UWB)*

INTRODUCTION

Since the release of a bandwidth of 7.5 GHz (from 3.1 to 10.6 GHz) for ultra-wideband (UWB) wireless communications by the Federal Communications Commission (FCC), antennas suitable for UWB systems have drawn much attention from researchers worldwide.

Due to their attractive merits (such as operating in an ultra-wide frequency range, near omnidirectional radiation patterns, simple structure, and low cost), considerable research effort has been devoted to broadband monopoles of various configurations [1–4] for UWB applications. However, they are not suitable for integra-

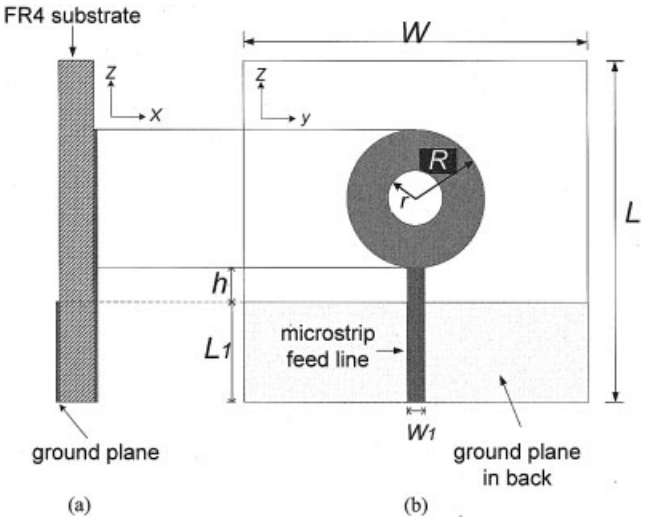


Figure 1 Geometry of the proposed printed circular disc/ring monopole: (a) side view; (b) top view

tion with a printed circuit board because their ground planes are perpendicular to the radiators. There is great demand for UWB antennas that offer a fully planar structure.

In this paper, we propose and investigate a novel design of microstrip line-fed circular-ring monopole based on our previous studies [5–7]. The design parameters for achieving optimal performance are investigated. It has been demonstrated both numerically and experimentally that the circular-ring monopole exhibits almost the same characteristics as the circular-disc monopole. The optimal design of this type of antenna can provide a return loss in excess of −10 dB over an extremely wide frequency range.

ANTENNA DESIGN

The geometry of the proposed microstripline-fed circular-ring monopole antenna is shown in Figure 1. The circular copper-ring radiator with an outer radius of $R = 10$ mm, an inner radius r, and the microstrip feed line are printed on the same side of the dielectric substrate (in this study, the FR4 substrate of thickness 1.5 mm and relative permittivity 4.7 was used). L and W denote the length and the width of the dielectric substrate, respectively. The width of the microstrip feed line is fixed at $W_1 = 2.6$ mm to achieve 50Ω impedance. On the other side of the substrate, the conducting ground plane with a length of $L_1 = 20$ mm only covers the section of the microstrip feed line. The feed gap h is the distance between the feed point of the antenna and the ground plane.

ANALYSIS AND DISCUSSION

All the simulations are performed using the CST Microwave Studio™ package, which utilizes the finite-integration technique for electromagnetic computation [8]. The simulations have shown that the performance of the microstripline-fed circular ring monopole is critically dependent on the feed gap h, the width of the ground plane W, and the inner radius of the ring r. The optimized h and W are found to be at $h = 0.3$ mm and $W = 42$ mm [7], respectively.

The simulated return-loss curves for different inner radius with $h = 0.3$ mm, $W = 42$ mm, and $L = 50$ mm are illustrated in Figure 2. It can be noted from Figure 2 that when r is no more than 4 mm, the return-loss curves do not change much with the variation of r. When r rises to 6 mm, the return loss shifts up to more than −10 dB at around 5 and 7 GHz. As a result, the −10-dB bandwidth is narrowed remarkably. This indicates that a circular-ring monopole with an inner radius of up to 4 mm can provide similar operating bandwidth as the counterpart circular-disc monopole. This is because the current is mainly distributed along the edge of the circular-disc monopole [6] and, hence, cutting the central part of the disc to form a ring monopole will not affect the performance of the monopole. It is also shown in Figure 2 that the first resonance always occurs at around 3.1 GHz for different r. This indicates that this resonance is mainly determined by the outer radius R, and not influenced by the inner radius r.

Figure 3 shows the simulated current distributions at different frequencies for the optimal design of the proposed ring monopole with $r = 4$ mm, $h = 0.3$ mm, $W = 42$ mm, and $L = 50$ mm. Figure 3(a) presents the current pattern near the first resonance at 3 GHz. The current pattern near the second resonance at 6.5 GHz is given in Figure 3(b), indicating an approximately 2nd-order harmonic; Figure 3(c) plots a more complicated current pattern at 9 GHz, corresponding to the 3rd-order harmonic. As shown in Figure 3, the current is mainly distributed along the outer edge of the circular ring for all of the three frequencies, like the circular-disc monopole [6].

A prototype of the proposed monopole antenna with optimal design, that is, $r = 4$ mm, $h = 0.3$ mm, $W = 42$ mm, and $L = 50$ mm, as shown in Figure 4, is fabricated and tested in the Antenna Laboratory at Queen Mary University of London (QMUL). The return losses are measured by using an HP 8720ES network analyzer and the radiation-pattern measurements were carried out inside an anechoic chamber.

Figure 5 illustrates the simulated and measured return-loss curves of the antenna. The measured return loss agrees very well with the simulated one in the whole frequency range from 0 to 12 GHz. The simulated −10-dB bandwidth ranges from 2.71 to 10.04 GHz, while the measured one covers a frequency band from 2.75 to 9.84 GHz, as shown in Figure 5.

The simulated and the measured radiation patterns at 3, 6.5, and 9 GHz are plotted in Figures 6–8, respectively. The measured patterns are close to those obtained in the simulation. The E-plane pattern is similar to that of a conventional monopole at lower frequency (3 GHz). At higher frequencies (6.5 and 9 GHz), it starts to form notches and become more directional at around ±30° from the z-direction. It is observed that the H-plane patterns are nearly omnidirectional over the entire bandwidth and their shapes correspond well to the current patterns on the circular ring, as shown in Figure 3.

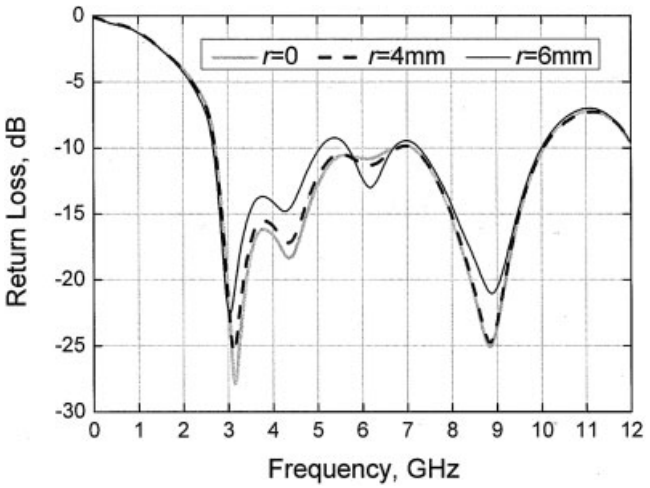


Figure 2 Simulated return-loss curves for different inner radii of the circular-ring monopole with $R = 10$ mm, $h = 0.3$ mm, $W = 42$ mm, and $L = 50$ mm

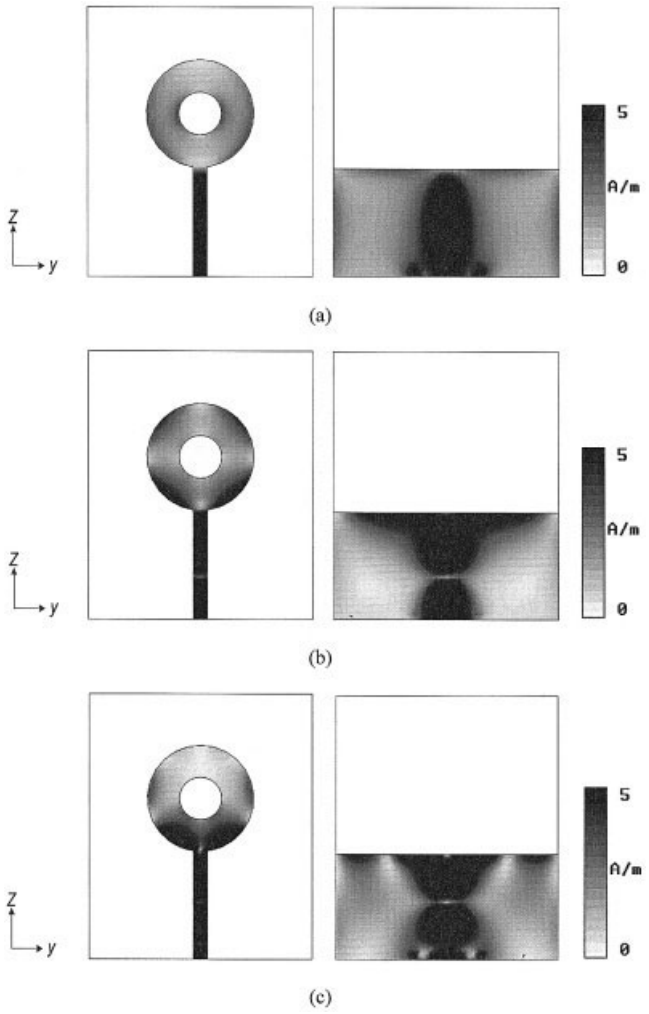


Figure 3 Simulated current distributions on the circular-ring monopole and the ground plane with $R = 10$ mm, $r = 4$ mm, $h = 0.3$ mm, $W = 42$ mm, and $L = 50$ mm at (a) 3 GHz, (b) 6.5 GHz, and (c) 9 GHz

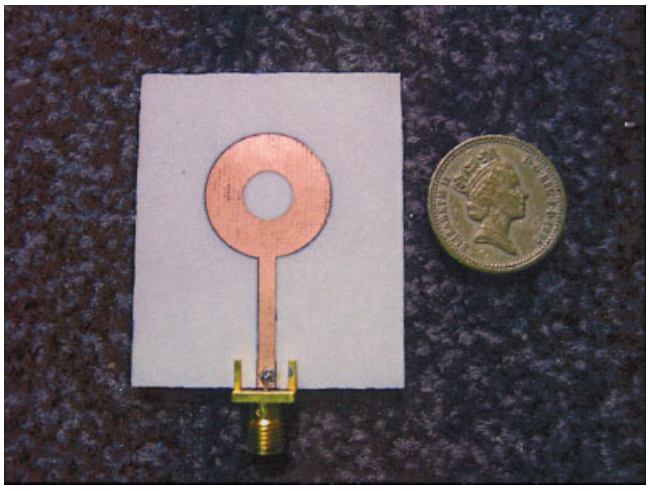


Figure 4 Photograph of the microstripline-fed circular-ring monopole. [Color figure can be viewed in the online issue, which is available at www.interscience.wiley.com.]

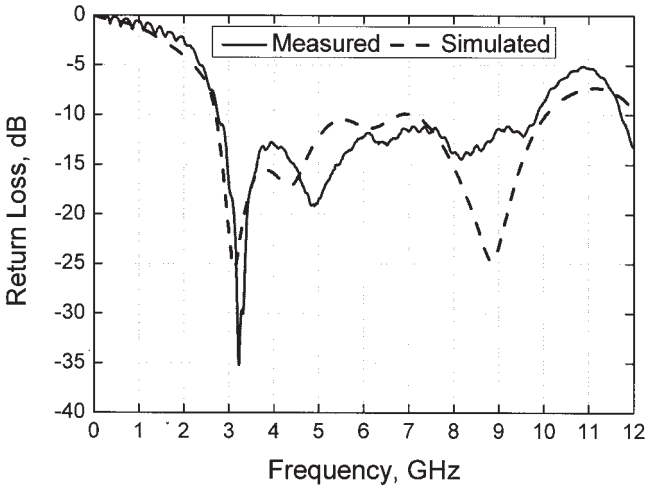


Figure 5 Simulated and measured return-loss curves of the microstripline-fed circular-ring monopole with $R = 10$ mm, $r = 4$ mm, $h = 0.3$ mm, $W = 42$ mm, and $L = 50$ mm

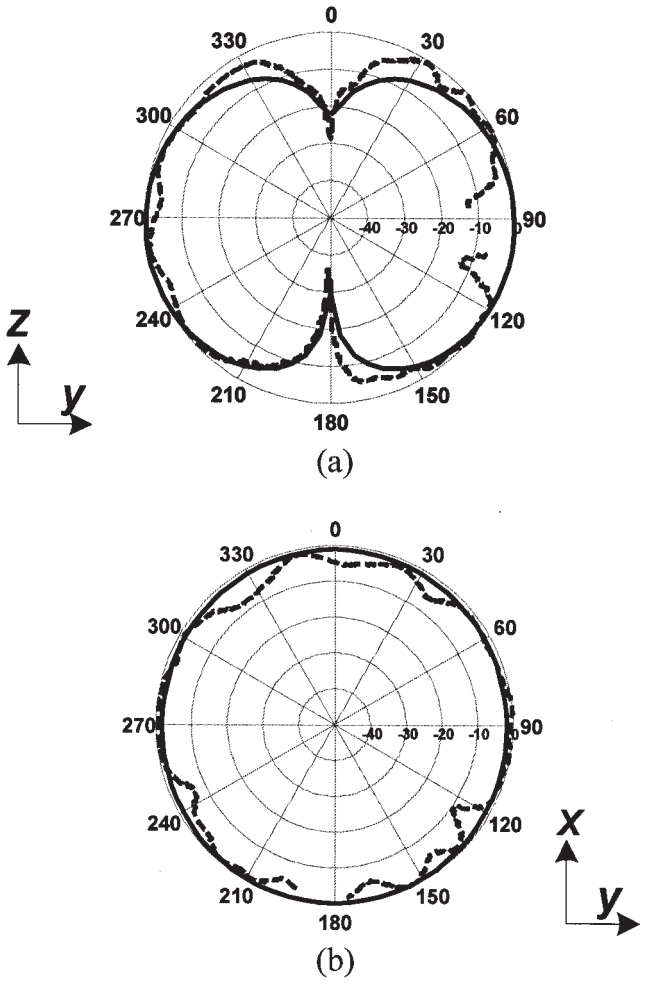


Figure 6 Simulated (solid line) and measured (dashed line) radiation patterns of the microstripline-fed circular-ring monopole with $R = 10$ mm, $r = 4$ mm, $h = 0.3$ mm, $W = 42$ mm, and $L = 50$ mm at 3 GHz: (a) E-plane; (b) H-plane

The measurements have demonstrated the UWB characteristic of the proposed antenna. It has also been confirmed that the circular-ring monopole can provide a performance similar to that of the circular-disc monopole [7], as predicted in the simulations.

CONCLUSION

A printed circular-ring monopole antenna fed by microstrip line has been proposed and investigated in this paper. It has been demonstrated numerically and experimentally that the circular-ring monopole exhibits characteristics similar to those of the circular-disc monopole because the current is mainly distributed along the outer edge of the ring. Both simulations and measurements have shown that the microstripline-fed circular-ring monopole can achieve ultra-wide operating bandwidth, covering the FCC-defined UWB frequency band. It is also observed that the radiation patterns of the proposed antenna are nearly omnidirectional over the entire operating bandwidth, similarly to those of a conventional monopole.

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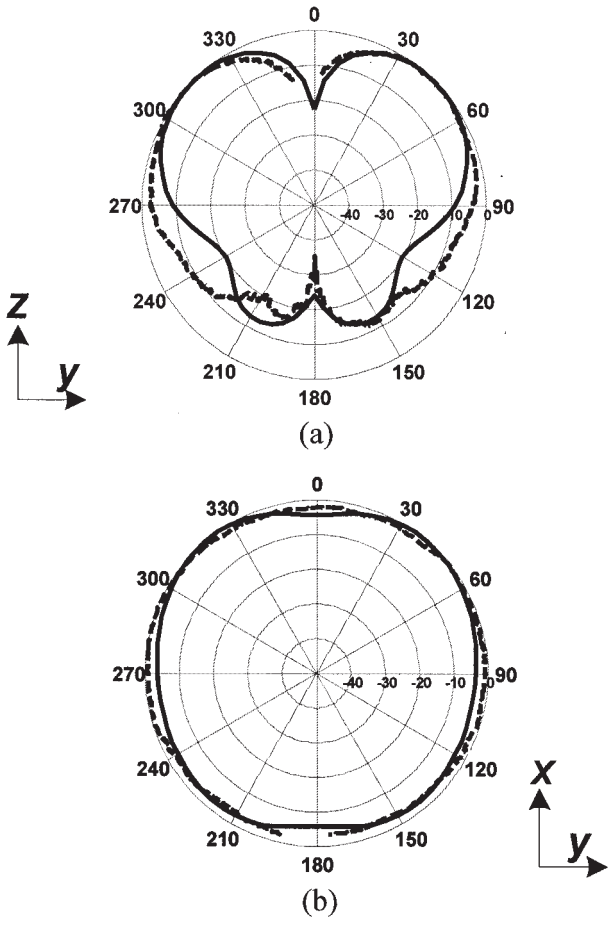


Figure 7 Simulated (solid line) and measured (dashed line) radiation patterns of the microstripline-fed circular-ring monopole with $R = 10$ mm, $r = 4$ mm, $h = 0.3$ mm, $W = 42$ mm, and $L = 50$ mm at 6.5 GHz: (a) E-plane; (b) H-plane

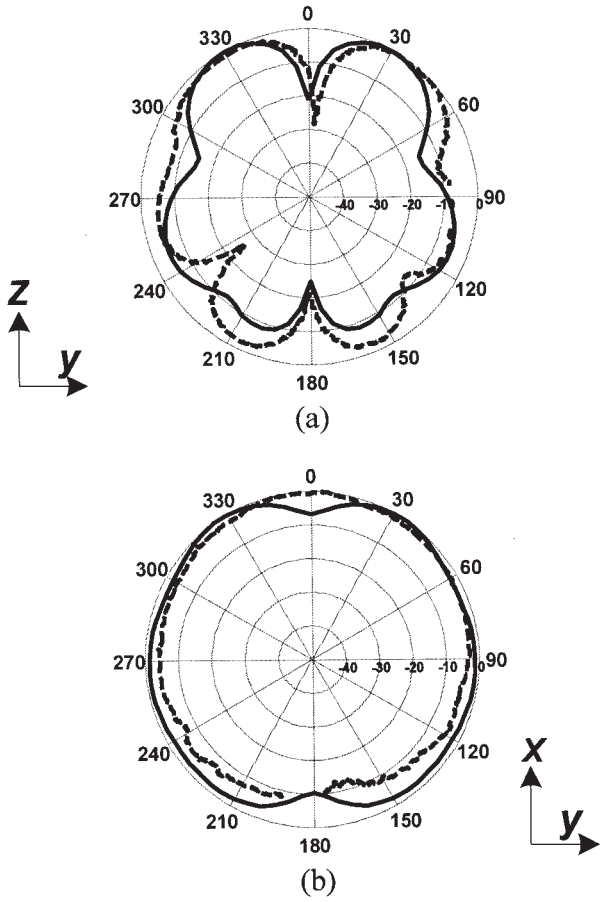


Figure 8 Simulated (solid line) and measured (dashed line) radiation patterns of the microstripline-fed circular-ring monopole with $R = 10$ mm, $r = 4$ mm, $h = 0.3$ mm, $W = 42$ mm, and $L = 50$ mm at 9 GHz: (a) E-plane; (b) H-plane

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